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EDC16U34 Disassembling

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EDC16U34 Disassembling

by **Szabi1** » Fri Jan 03, 2025 12:35 am

Hello everyone! I have a 03G906021AB HW07 hardware from Passat BMP and I want to disassembly the code. Unfortunatelly my BDM100 tool only can read flash and eeprom in this ECU family, I have tried with EDC16U31 where it can read the MPC processor firmware as well. Does somebody have a MPC firmware read from this EDC16u34 ecu family with MPC562 or does it has at all?
Right now this ecu lives in a MK5 jetta BKC. The main reason for ecu swap it was to add an EGT sensor in oem style to a non-dpf car from factory. Unfortunatelly I haven't succeeded to turn off EBP stuff in passat flash without stuck in bootloop, so I moved to Audi A3 BMM software written via BDM into the unit. That hardware is closer to MK5 jetta anyways (yes calibration data was adapted ("maps" which is tweeked by tuners usually) and running fine).

So back to the hardware: If I turn on the sensor driver in the ori flash or I make BDM copy from the Audi the ECU boots up sucessfully but throws dtc about the egt sensor input is short to ground, so is not the issue in the flash or eeprom content.
I am not sure that does the hardware differs by unused pins are grounded on the older non dpf units or the I/O pin declarations is inside the MPC firmware? If we are talking about the EDC15 family others confirmed that the "irom" cpu firmware is the same in both EDC15VM and EDC15P+ so the chances are that it's probably the same here for both EDC16, but unfortunately I don't know how to read it out....

I am also curious about other things in the code, like how ACC works, etc. so it would be nice if it can be disassembled somehow.

Szabi1
Newbie

Posts: 93
Joined: Sat Jan 29, 2022 1:12 pm
Has thanked: 14 times
Been thanked: 33 times
Contact:

Re: EDC16U34 Disassembling

Szabi1

by **Szabi1** » Fri Jan 03, 2025 3:08 am

Sorry I'm so dumb, that I haven't started with the CPU documentation. Now is clear that the MPC562 doesn't have any internal ROM (firmware) inside.

Newbie

Posts: 93
Joined: Sat Jan 29, 2022 1:12 pm
Has thanked: 14 times
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Re: EDC16U34 Disassembling

by **eliaskotlyar** » Tue Jan 21, 2025 1:21 pm

I tried disasm a EDC16 once with ghidra, it got very complicated. It uses not only the two globals(R2,R13) but also some pointer table where all the map-addresses very held. Ghidra barely could find some reference. I stopped doing it as i got stuck.

I can provide my ghidra project if it helps somehow...

eliaskotlyar
Newbie

Posts: 14
Joined: Thu Nov 28, 2024 2:46 pm
Has thanked: 1 time
Been thanked: 3 times
Contact:

Re: EDC16U34 Disassembling

by **trizex** » Tue Jan 21, 2025 7:11 pm

have successfully disassembled (and modified - wrote custom code) first bmw's whole edc16 range, and just recently have managed to do the same for edc16u1 /u31 /u34. Will probably expand the range of ecus more when i have time for it

As Elias pointed out above, its a enormous endeavour, I've spend literal hundreds of work hours for learning/understanding first edc15/c166 and more recently edc16/mpc, so I wont be sharing my knowledge for free, but they will soon be for sale - have done a few bmws already.

Sneak peek attached below, m57n2 286 switching on/off antilag/pNb via DTC button

https://youtube.com/shorts/LcpUwrkuUug

These users thanked the author trizex for the post:
supertramp78

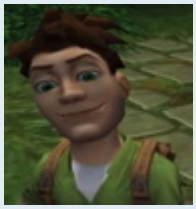
trizex
Stage 1

Posts: 115
Joined: Thu Sep 28, 2017 11:00 am
Has thanked: 44 times
Been thanked: 126 times
Contact:

Re: EDC16U34 Disassembling

by **Danielsvamp** » Thu Feb 06, 2025 3:11 am

> **trizex wrote:** Tue Jan 21, 2025 7:11 pm
> have successfully disassembled (and modified - wrote custom code) first bmw's whole edc16 range, and just recently have managed to do the same for edc16u1 /u31 /u34. Will probably expand the range of ecus more when i have time for it
>
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>
> Sneak peek attached below, m57n2 286 switching on/off antilag/pNb via DTC button
>
> https://youtube.com/shorts/LcpUwrkuUug



Danielsvamp
Newbie

Posts: 19
Joined: Fri Dec 23, 2022 11:30 pm
Has thanked: 36 times
Been thanked: 1 time
Contact:

That's amazing work, keep it up! Dreaming of learning this myself, easily worth hundreds of hours

Re: EDC16U34 Disassembling

by **Djang01** » Thu Nov 13, 2025 10:01 pm

> **Szabi1 wrote:** Fri Jan 03, 2025 12:35 am
> Hello everyone! I have a 03G906021AB HW07 hardware from Passat BMP and I want to disassembly the code. Unfortunatelly my BDM100 tool only can read flash and eeprom in this ECU family, I have tried with EDC16U31 where it can read the MPC processor firmware as well. Does somebody have a MPC firmware read from this EDC16u34 ecu family with MPC562 or does it has at all?
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>
> So back to the hardware: If I turn on the sensor driver in the ori flash or I make BDM copy from the Audi the ECU boots up sucessfully but throws dtc about the egt sensor input is short to ground, so is not the issue in the flash or eeprom content.
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>
> I am also curious about other things in the code, like how ACC works, etc. so it would be nice if it can be disassembled somehow.

Non DPF ECU has pin pointing to nothing inside, i tried the same thing as you, then opened up an ecu and traced the pin inside

Djang01
Newbie

Posts: 17
Joined: Tue Jun 30, 2015 5:57 pm
Has thanked: 1 time
Been thanked: 1 time
Contact:

Re: EDC16U34 Disassembling

by **rodolfo** » Sun Nov 16, 2025 12:05 am

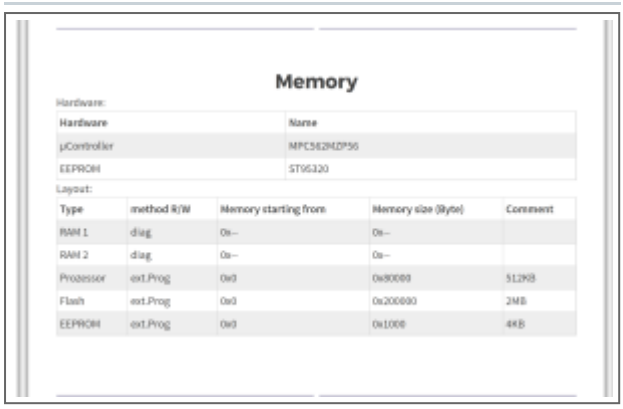
May be it can help you

ATTACHMENTS



rodolfo
Newbie

Posts: 26
Joined: Sat Mar 08, 2014 11:37 am
Has thanked: 7 times
Been thanked: 10 times
Contact:



Memory

Hardware:

Hardware	Name
µController	MPC5612MZP56
EEPROM	ST95320

Layout:

Type	method R/W	Memory starting from	Memory size (Byte)	Comment
RAM 1	diag	0x--	0x--	
RAM 2	diag	0x--	0x--	
Prozessor	ext.Prog	0x0	0x80000	512KB
Flash	ext.Prog	0x0	0x200000	2MB
EEPROM	ext.Prog	0x0	0x1000	4KB

Re: EDC16U34 Disassembling

by **Szabi1** » Sun Nov 16, 2025 11:05 pm

Here is what I've figured out, specifically for the EDC16U34, based on leaked documents found on various forums. There is a so-called "**SW_Basis**" which I believe is developed by the ECU manufacturer — in this case, BOSCH. In the following example, the SW_Basis is fixed to "**C447HAXN**" This version string always embedded into an OBD read as well.
The internal logic of the software is documented by BOSCH. Every map and every map axis has an official ID (for example: for duration map 5, InjVlv_phiInjMI1_MAP5; for the RPM axis, Eng_nAvrg; for the IQ axis, InjUn_qMI1DesCorr; and so on). In the "Dokumente" folder you can find the software release information for this specific SW_Basis.

The software/function documentation is sometimes referred to as the "Funktionsrahmen." We don't have a leaked PDF for this exact SW_Basis version, but we do have one for **HAXL**, as well as "delta documentation" for **HAXM** and **HAXN** (the software versioning follows alphabetical order). We also have the "Dokumente" folder for each of these versions, including their changelogs.

Next, we have what is usually called the "real damos" or ASAP2/A2L files. These contain all calibration data addresses in flash and in RAM; we also refer to this as a mappack. I previously uploaded one for the Passat BMP, which matches with this software version. It is superior to a WinOLS KP file — for example, if you resize a 2D map, it can still detect the map's location. I believe this file is generated by BOSCH and used by the car manufacturer's engineers to fine-tune the engine and the vehicle.
There is also something Volkswagen calls "System-spezifische Software-Daten," or SSD for short. These are text files that contain some calibration data for a specific vehicle platform such as PQ35 and for a specific engine type (displacement, number of valves, DPF or non-DPF, etc.).

For an ECU target (for example 03G906021NK), a separate flash file was generated for every "variant coding," which is also referred to as a variant file. These variant files contain the SW_Basis plus the corresponding calibration data (dataset), and they are encoded in Intel HEX format (e.g. 1037399388_var_1.hex or HAXNCHHQ4000.hex). There are tools that can decode them into raw binary — which I did — and all bits end up in the exact same positions as in a BDM flash read. Usually, datasets contain different calibration and configuration data depending on the transmission type.
For this specific ECU target, for example, three such variant files were generated, each containing the SW_Basis and one calibration. The first one I think is for the automatic transmission, the second for the 5-speed manual,

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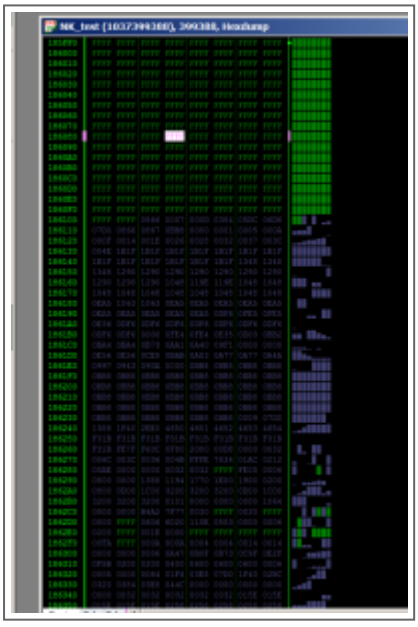
and the third for the 6-speed manual. In other words, one dataset in a variant file for EDC16 corresponds roughly to what a codeblock was in EDC15.
If you convert any of these Intel HEX files to BIN (which I did), I think you can simply flash them to the ECU — as long as the hardware matches — and the ECU will run with them.

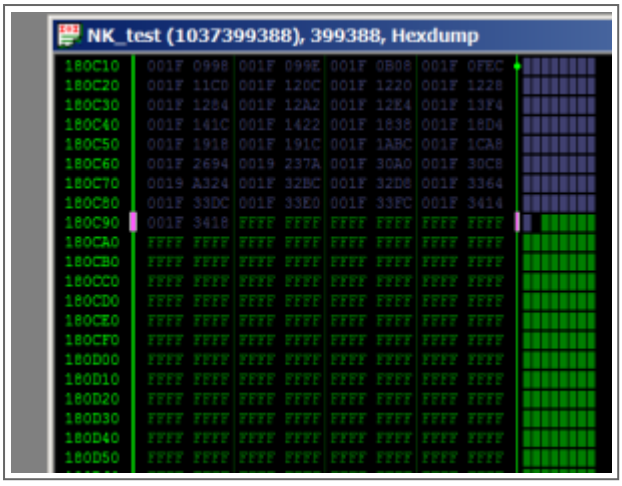
However, most likely to save space, not every "codeblock" contains a full copy of every calibration map as in EDC15. What I observed is that between the different datasets, only the maps that actually change are included again in the final IAV delivery's Variant Data area at: 0x180000 - 0x1BFFFF: 0x6871 (You can check the images I've included)
So the "IAV delivery" content is 1:1 what we have in the flash read by BDM on an original untouched ECU.

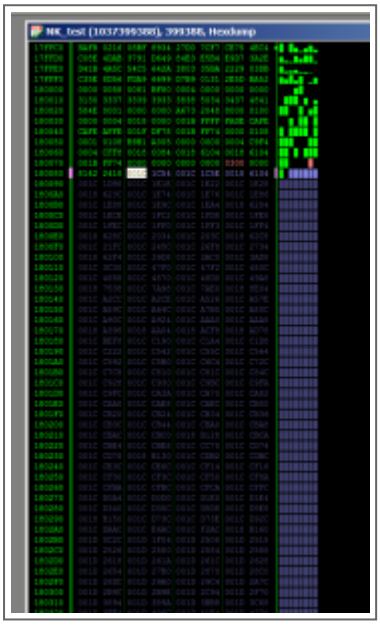
So if we can figure out how the variant data area is configured to act as links or references to the main calibration maps, we could theoretically create unlimited “codeblocks” from which we can try to implement multimap logic similar to EDC15.
Please give me a thumbs up if you find this discovery useful — peace 😉

One more thing: I've tried to patch the manual transmission dataset into one for the automatic in my ecu, but after changeing the coding in vcds the ABS thrown fault that no comm with the transmission ecu, but the ECU does not had any error codes. Maybe I've missed out something.

ATTACHMENTS







These users thanked the author Szabi1 for the post:
DerGeraet

Re: EDC16U34 Disassembling

by **eliaskotlyar** » Thu Nov 27, 2025 2:56 am

Post your files (a2l + matching bin/hex) and i can try to make you a ghidra project. i have some sophisticaed homemade tools for such adventures and hopefully it will help you on your journey.

For figuring out the memory-map, you can also use VAG update files and a tool called SGO extractor. It will display the memory mapping of the update file while extracting the SGO into a bin.

eliaskotlyar
Newbie

Posts: 14
Joined: Thu Nov 28, 2024 2:46 pm
Has thanked: 1 time
Been thanked: 3 times
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